A Sequential Approach for Sentinel-1 TOPS Time-Series Co-Registration Over Low Coherence Scenarios

Zhang-Feng Ma, *Student Member, IEEE*, Mi Jiang, *Senior Member, IEEE*, and Teng Huang

***Abstract*— In the coming era of synthetic aperture radar (SAR) big data, the in-orbit Sentinel-1 mission will provide unprecedented data with an increasing volume. As a fundamental step of time-series analysis for such large and growing amount data, terrain observation by progressive scans (TOPS) co-registration still presents a relative challenge: 1) low coherence scenarios may degrade the estimate accuracy and 2) unprecedented and growing data volume increases the computational burden. To overcome both limitations, this article presents a sequential approach for TOPS time-series co-registration, with an emphasis on the enhanced spectral diversity (ESD) estimate accuracy over low coherence scenes. We first employ double sample over the burst overlap region to improve the statistical proprieties of sample covariance matrix, followed by ESD phase estimation using a phase linking algorithm. Then, we carry out the sequential co-registration on each mini-stack without the necessity for reprocessing the entire stack by introducing a data compression technique. Using synthetic data and real Sentinel-1 TOPS data over densely vegetated areas in the Yunan–Kweichow plateau, we fully evaluate the performance of presented approach and compare the results with those obtained from the state-of-the-art techniques. We found that the sequential approach can provide better time-series co-registration accuracy over low coherence scenes with the moderate computational efficiency.**

***Index Terms*— Co-registration, eigen decomposition, enhanced spectral diversity (ESD), sequential estimator, synthetic aperture radar (SAR) interferometry (InSAR), terrain observation by progressive scans (TOPS).**

I. Introduction

THE Sentinel-1 synthetic aperture radar (SAR) mission can provide dense time-series SAR data to enable a systematic and continuous monitoring of the Earth's surface [2]. In this context, terrain observations by progressive scans (TOPS) [3], the default imaging SAR mode of interferometric wide swath (IW), can capture three subswaths with a coverage of 250 km by the steering antenna. In contrast to ScanSAR mode [4], TOPS mode can achieve nearly uniform signal-to-noise ratio (SNR) and the distributed target ambiguity ratios. Nevertheless, the steering of the antenna will result in a large azimuth-dependent Doppler centroid variation, which introduces the phase ramps in differential interferograms when the azimuth co-registration error exists. Given an acceptable phase ramp of 1/100 cycle (∼3.6°) at a Doppler centroid difference of 5000 Hz, the theoretical co-registration accuracy of ∼0.001 pixels in the azimuth direction is required for Sentinel-1 TOPS interferometric applications. This is generally achieved by a two-step strategy [5].

In the first step, a geometrical approach [6], [7] uses the precise orbit and an external digital elevation model (DEM) to match all images with respect to a common reference. Because the Sentinel-1 precise orbit has a 3-D 1 sigma accuracy of ∼5 cm [8], a relatively high co-registration accuracy in both range and azimuth direction can be obtained. In scenarios with only cross-track motion, a deterministic orbit maintenance of a ∼120-m orbital tube ensures the azimuth mis-registration errors topographically independent and turn into a constant [8]. However, when regional ionospheric delay [9], glacier motion [10], and seismic events [11] appear, the along-track signals may contribute to phase inconsistency in consecutive burst interferograms and complicate azimuth co-registration problem in the second step [9], [12].

Given a constant azimuth misregistration error, the range shift after geometrical co-registration can be further refined by the cross-correlation algorithm and a polynomial warp function [10], [13]. In the azimuth direction, two techniques based on azimuth spectral properties are commonly applied: spectral diversity (SD) and enhanced spectral diversity (ESD) [13], [14]. SD is essentially consistent with the split bandwidth technique [15] and multiaperture interferometry (MAI) [16] in spite of different nomenclatures. These techniques differentiate the two sublooks in azimuth Doppler spectrum to obtain the differential interferogram, in which each target presents a constant phase proportional to the misregistration error and the Doppler variation within the mainlobe of impulse response function (IRF). In contrast, ESD uses the spectral information of pixels in the overlap region of consecutive bursts, which is captured by different squinted angles. This technique has a much higher spectral separation (∼5000 Hz) than azimuth bandwidth (∼320 Hz) of

Manuscript received November 1, 2019; revised February 19, 2020 and June 22, 2020; accepted July 13, 2020. Date of publication July 29, 2020; date of current version May 21, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 41774003, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20171432, and in part by the ESA-MOST Dragon 4 Project under Grant 32248_2. *(Corresponding author: Mi Jiang.)*

Zhang-Feng Ma and Teng Huang are with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China (e-mail: jspcmazhangfeng@hhu.edu.cn; njht0911@163.com).

Mi Jiang is with the State Key Laboratory of Hydrology-Water Resources and Hydraulic Engineering, Hohai University, Nanjing 210098, China, and also with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China (e-mail: mijiang@hhu.edu.cn).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3009996

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF DIFFERENT METHODS

Method	Temporal Network	Least Squares	Phase Optimization	Coherence Optimization
NESD	Multi-Master	Yes	Multi-looking	No
MST-ESD	MST	Yes	No	Yes
Joint Estimator	Multi-Master	Yes	PL	Yes
EESD	Single-Master	No	Refined EMI + Sequential Estimator	Yes

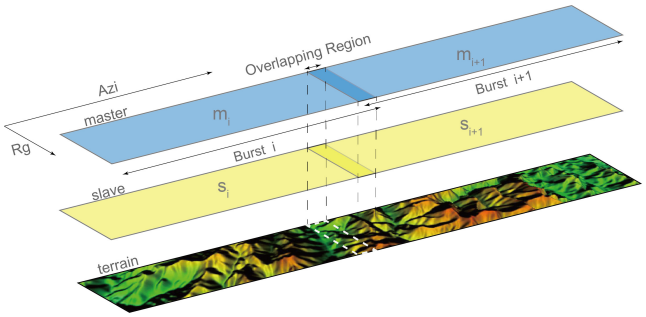


Fig. 1. General description of ESD.

SD, resulting in a higher sensitivity to azimuth misregistration errors [14].

Because ESD uses the interferometric phase to determine the azimuth shift, the estimate accuracy of ESD relies on coherence. When time-series data are available, the abrupt loss of coherence between two arbitrary acquisitions over low coherence scenarios may bias the co-registration performance.

To avoid coherence loss, several efforts have been made in the past five years under graph theory or data statistics. For example, the network-based ESD (NESD) method [9] connected neighboring pairs to avoid the temporal decorrelation, followed by a weighted least-squares adjustment (Table I). MST-ESD employed a minimum spanning tree (MST) [17] to maximize coherence and simultaneously mitigate the influence of biased coherence estimator on ESD estimation. Different from NESD and MST-ESD that highlight coherence, the joint ESD estimator [18] enhanced the phase qualities for an SAR stack by the phase-linking (PL) [19] technique, and then applied a variance–covariance matrix for a least squares inversion [20]. Generally, these algorithms have performed well in most applications but cannot balance the estimation accuracy and computational efficiency. Moreover, a sequential TOPS co-registration scheme to avoid rereading and reprocessing the entire data volume when adding a new acquisition is rarely discussed.

This article aims to improve the ESD estimate accuracy over low coherence areas and to mitigate the computational burden for big SAR data co-registration. To this end, we deploy the eigen decomposition-based maximum-likelihood estimator of interferometric phase (EMI) [21] and double sample in the overlap region to improve the ESD estimation efficiency. The motivation behind this method is that double sample improves the statistic properties of covariance matrix, whereas EMI can accelerate the phase optimization. Motivated in [1], we further use a data compression technique to divide the stack into isolated mini-stacks and then retrieve the coherent signal between mini-stacks without the necessity for reprocessing the entire stack.

The rest of this article is organized as follows. Section II presents our methodology. The effectiveness of presented method versus the state-of-the-art techniques are fully evaluated and compared by synthetic and real data sets in Sections III and IV, respectively. Finally, conclusions are outlined in Section V.

II. METHODOLOGY

A. Enhanced Spectral Diversity

After the geometrical co-registration [6], the azimuth residual ESD phase ϕ_{ESD} caused by the systematical orbital uncertainty can be estimated by [13]

$$\phi_{\text{ESD}} = \angle\left(\left(m_i \cdot s_i^*\right) \cdot \left(m_{i+1} \cdot s_{i+1}^*\right)^*\right) \tag{1}$$

where m_i and s_i are the ith master and slave burst, respectively (Fig. 1), $m_i \cdot s_i^*$ means burst interferogram.

Theoretically, the azimuth shift $\widehat{\Delta x}$ can be estimated from the wrapped phase histogram using the periodogram [18], [22]

$$\widehat{\Delta x} = \arg\max_{\Delta x}\left\{\left|\Re\left(\sum_p e^{j\left(\phi_{\text{ESD},p} - 2\pi\,\Delta f_{\text{ovl}}\tau\,\Delta x\right)}\right)\right|\right\} \tag{2}$$

where $\phi_{\text{ESD},p}$ represents the residual ESD phase of a coherent pixel p, Δf_{ovl} is the focused Doppler frequency difference between consecutive bursts, τ denotes the line time interval, and $\Re$ denotes the real part. The accuracy $\sigma_{\Delta x}$ can be given as [9], [13]

$$\sigma_{\Delta x} = \frac{1}{2\sqrt{n}\pi\,\Delta f_{\text{ovl}}\tau}\frac{\sqrt{1-\gamma^2}}{\gamma} \tag{3}$$

where γ is the coherence and n denotes the number of samples used to average.

B. Application of EMI Over Burst Overlaps

Compared with the PL technique used in previous studies [18], [19], EMI algorithm is proven to be a more effective way to improve the qualities of phase series and computational costs [21]. The mathematical formulation of EMI is defined as [21]

$$\begin{cases} \hat{\Sigma} = \dfrac{1}{n}\mathbf{g}\mathbf{g}^{\text{H}} \\ \hat{\theta} = \arg\min\limits_{\theta}\left\{\xi^{\text{H}}\left(\hat{\Gamma}^{-1} \circ \hat{\Sigma}\right)\xi\right\} \end{cases} \tag{4}$$

where $\hat{\theta} = \angle\xi^{-1}$ is optimum time-series phase and ξ is the minimum eigen-pair of the Hadamard product $\hat{\Gamma}^{-1} \circ \hat{\Sigma}$; the superscript H represents the Hermitian transposition; $\mathbf{g} \in \mathbb{C}^{m\times n}$ is an $m \times n$ observation matrix of a reference pixel, in which the row denotes the temporal single look complex (SLC) sample g and the column denotes the spatial samples in a homogenous region $\boldsymbol{\Omega}$ that is determined by the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

FaSHPS algorithm [23]; $\hat{\Sigma}$ represents the covariance matrix and $\hat{\Gamma}^{-1}$ is the inverse of the coherence matrix $\hat{\Gamma}$, in which each element is coherence between two arbitrary acquisitions i and j

$$\left\{\hat{\Gamma}\right\}_{i,j} = \frac{1}{n}\left|\sum_{p\in\Omega}\exp\left(1j\left(\angle g_{i,p} - \angle g_{j,p} - \phi_p^{\text{ramp}}\right)\right)\right| \tag{5}$$

where ϕ^{ramp} represents a linear phase compensation used to remove locally nonstationary signals such as deformation pattern [24]. In (5), the amplitude in both SLC images is ignored to avoid any unbalance between SAR amplitude [25]. This behavior helps to eliminate the overestimation of coherence due to the appearance of a bright pixel in a dark area.

EMI requires the assumption of sufficiency of the sample coherence matrix close to truth [21]. Otherwise, the coherence error will be amplified by the inverse operator [21], [25]. This assumption, however, is difficult to be fulfilled due to the limited ensembles and the biased estimator [25], [26]. A simple but very efficient way to improve the estimate accuracy of sample coherence matrix is to use double sample [22], i.e., mitigating the coherence error by increasing the sample size at two consecutive burst overlaps under the assumption that the targets in the upper and lower overlap regions in Fig. 1 share the similar decorrelation mechanism during the imaging time interval. Theoretically, the increase of sample size can simultaneously reduce the bias and variance of sample coherence [24], resulting in improved estimate accuracy for any nondiagonal element in (5). The estimator (5) can be reformulated as follows:

$$\left\{\hat{\Gamma}\right\}_{i,j} = \frac{1}{2n}\left|\sum_{k=1}^{2}\sum_{p\in\Omega}\exp\left(1j\left(\angle g_{i,p}^{k} - \angle g_{j,p}^{k} - \phi_p^{\text{ramp}_k}\right)\right)\right| \tag{6}$$

where $k = 1, 2$ corresponds to ith and $i+1$th burst in Fig. 1, respectively. In addition, because the sample size in (6) is double than in (5), the positive definite assumption is also likely to be satisfied, i.e., the number of samples used for coherence matrix estimation should be larger than the stack size [27].

The performance of applying double sample into EMI has been fully evaluated in Sections II-C and III. We should highlight that the mathematic formulation of EMI estimator is not changed in this article. The method presented is hereafter referred to as refined EMI.

C. Application of Sequential Estimator in ESD

In this section, a sequential estimator developed from [1] for time-series analysis is introduced to estimate the azimuth misregistration.

Suppose a stack of m SLCs after geometrical co-registration, sequential estimator first divides this stack into K mini-stacks with a constant size s and a remainder after division with size $m - K \times s$.

In order to better describe the phase estimation and data compression in ESD, several notations should be introduced [1].

TABLE II

PSEUDOCODE OF APPLICATION OF SEQUENTIAL ESTIMATOR FOR ESD ESTIMATION IN A BURST OVERLAP AREA

1	Divide m images to K mini-stacks with size of s. Reminder $m - K \times s$ images after modulo are also treated as a mini-stack. The number of mini-stacks is K+1.
2	***for*** k = 1 *to* K+1
3	***if*** k = 1, **Input data:** $\hat{\mathbf{g}}_i^k = \mathbf{g}_i^k$, $\hat{\mathbf{g}}_{i+1}^k = \mathbf{g}_{i+1}^k$
4	***else,*** **Input data:** $\hat{\mathbf{g}}_i^k = [\tilde{g}_i^1;...,\tilde{g}_i^{k-1};\mathbf{g}_i^k]$, $\hat{\mathbf{g}}_{i+1}^k = [\tilde{g}_{i+1}^1;...,\tilde{g}_{i+1}^{k-1};\mathbf{g}_{i+1}^k]$
5	***end***
6	**Refined EMI Eq. (4)** $\rightarrow$ $\hat{\theta}_i^k$, $\hat{\theta}_{i+1}^k$
7	**Isolated mini-stack Compression (Fig.2b)** $\rightarrow$ $\hat{z}_{cal,i}^k$, $\hat{z}_{cal,i+1}^k$
8	***if*** k^{th} mini-stack size = s
9	**Merged set Compression** $\rightarrow$ $\tilde{g}_i^k$, $\tilde{g}_{i+1}^k$
10	**Data Archiving:** $\tilde{g}_i^k$, $\tilde{g}_{i+1}^k$, $\hat{z}_{cal,i}^k$, $\hat{z}_{cal,i+1}^k$
11	***end***
12	***end***
13	**Phase Connection:** $[\hat{\theta}_{cal,i}^1,\hat{\theta}_{cal,i}^2,...,\hat{\theta}_{cal,i}^{K+1}]$, $[\hat{\theta}_{cal,i+1}^1,\hat{\theta}_{cal,i+1}^2,...,\hat{\theta}_{cal,i+1}^{K+1}]$
14	**Connected Phase Series:** $[\hat{\theta}_i^1,\hat{\theta}_i^2,...,\hat{\theta}_i^{K+1}]$, $[\hat{\theta}_{i+1}^1,\hat{\theta}_{i+1}^2,...,\hat{\theta}_{i+1}^{K+1}]$
15	**ESD Phase Series: Differencing Connected Phase Series by Eq.(1)**
16	**Mis-registration Estimation Eq. (2)** $\rightarrow$ $[\widehat{\Delta x}^1,\widehat{\Delta x}^2,...,\widehat{\Delta x}^{K+1}]$

1) $\mathbf{g}_i^k, \mathbf{g}_{i+1}^k \in \mathbb{C}^{x\times n}$: burst overlap SLCs of the kth isolated mini-stack (containing n homogenous pixels) where i and $i+1$ mean the index of ith and $i+1$th burst overlap SLCs, respectively. When $k \le K$, $x = s$, and when $k = K+1$, $x = m - K \times s$.
2) $\hat{\mathbf{g}}_i^k, \hat{\mathbf{g}}_{i+1}^k \in \mathbb{C}^{(k+x-1)\times n}$: the kth merged set.
3) $\tilde{g}_i^k, \tilde{g}_{i+1}^k \in \mathbb{C}^{1\times n}$: the kth compressed merged set.
4) $\hat{\theta}_i^k, \hat{\theta}_{i+1}^k \in \mathbb{R}^{x\times 1}$: the optimum phase series of $\hat{\mathbf{g}}_i^k$ and $\hat{\mathbf{g}}_{i+1}^k$.
5) $\hat{z}_{\text{cal},i}^k, \hat{z}_{\text{cal},i+1}^k \in \mathbb{C}^{1\times n}$: the kth compressed isolated mini-stack.
6) $\hat{\theta}_{\text{cal},i}^k, \hat{\theta}_{\text{cal},i+1}^k \in \mathbb{R}^{1\times 1}$: the calibration phase of kth isolated mini-stack.

In Table II, input data are defined as the combination of newly added isolated mini-stack g_i^k, g_{i+1}^k and the previous compressed merged set $[\tilde{g}_i^1;\ldots,\tilde{g}_i^{k-1}]$ and $[\tilde{g}_{i+1}^1;\ldots,\tilde{g}_{i+1}^{k-1}]$. Then, the covariance matrix is generated using the merged set $\hat{\mathbf{g}}_i^k$. In data compression step, principal component analysis (PCA) [28] is applied to identify the dominant scattering mechanism with the largest eigen-pair. This eigen-pair represents the best (in the Frobenius norm sense) approximation of the covariance matrix by a single elementary scattering mechanism [29], [30] and reduces the data volume by retaining the main information. For example, assuming a reference pixel with stack size $m = 30$ and homogeneous pixel number $n = 200$: $\mathbf{g}_i, \mathbf{g}_{i+1} \in \mathbb{C}^{30\times 200}$, optimal phase series $\hat{\theta}_i^1, \hat{\theta}_{i+1}^1 \in \mathbb{R}^{10\times 1}$ of the isolated mini-stack $\mathbf{g}_i^1, \mathbf{g}_{i+1}^1 \in \mathbb{C}^{10\times 200}$ are generated by performing refined EMI using (4). Then, the first isolated mini-stack $\mathbf{g}_i^1, \mathbf{g}_{i+1}^1$ is compressed to vectors $\tilde{g}_i^1, \tilde{g}_{i+1}^1 \in \mathbb{C}^{1\times 200}$ using the PCA technique. Next, the previous compressed vectors $\tilde{g}_i^1, \tilde{g}_{i+1}^1$ and the second mini-stack $\mathbf{g}_i^2, \mathbf{g}_{i+1}^2 \in \mathbb{C}^{10\times 200}$ are combined to $\hat{\mathbf{g}}_i^2, \hat{\mathbf{g}}_{i+1}^2 \in \mathbb{C}^{11\times 200}$. $\hat{\theta}_i^2, \hat{\theta}_{i+1}^2 \in \mathbb{R}^{10\times 1}$ are obtained by refined EMI on $\hat{\mathbf{g}}_i^2, \hat{\mathbf{g}}_{i+1}^2$ (phases of $\tilde{g}_i^1, \tilde{g}_{i+1}^1$ are removed from $\hat{\theta}_i^2, \hat{\theta}_{i+1}^2$). Then, merged set $\hat{\mathbf{g}}_i^2, \hat{\mathbf{g}}_{i+1}^2$ are compressed as vectors $\tilde{g}_i^2, \tilde{g}_{i+1}^2 \in \mathbb{C}^{1\times 200}$, respectively. Finally, $\hat{\mathbf{g}}_i^3, \hat{\mathbf{g}}_{i+1}^3 \in \mathbb{C}^{12\times 200}$ are formed by combining the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

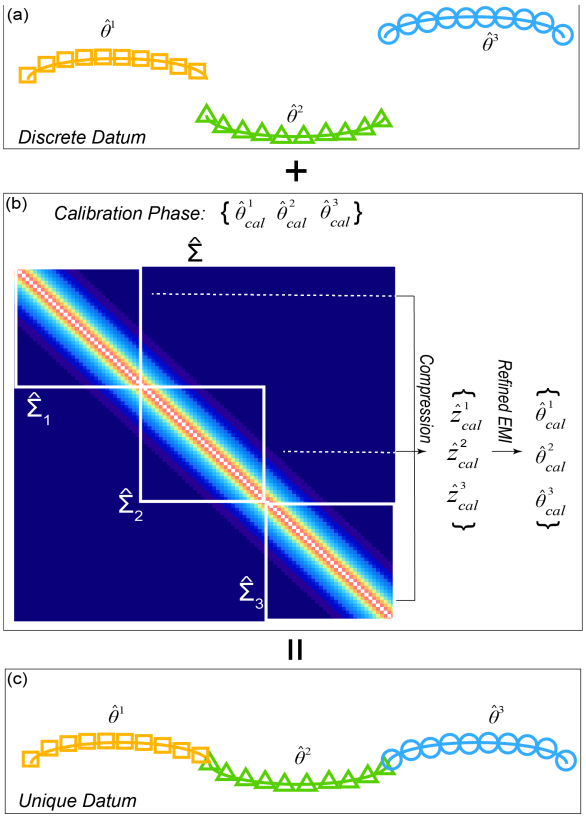


Fig. 2. Schematic description of the phase connection for single burst SLC series in the overlap area. (a) Discrete phase datums. (b) Calibration phases. (c) Unique datums.

previous compressed vectors and the newly added mini-stack: $[\tilde{g}_i^1; \tilde{g}_i^2; \mathbf{g}_i^3]$ and $[\tilde{g}_{i+1}^1; \tilde{g}_{i+1}^2; \mathbf{g}_{i+1}^3]$. Once $\hat{\theta}_i^3, \hat{\theta}_{i+1}^3 \in \mathbb{R}^{10\times 1}$ and $\tilde{g}_i^3, \tilde{g}_{i+1}^3 \in \mathbb{C}^{1\times 200}$ are obtained, three isolated mini-stacks are individually compressed to $\hat{z}_{\text{cal},i}^1, \hat{z}_{\text{cal},i}^2, \hat{z}_{\text{cal},i}^3 \in \mathbb{C}^{1\times 200}$ and $\hat{z}_{\text{cal},i+1}^1, \hat{z}_{\text{cal},i+1}^2, \hat{z}_{\text{cal},i+1}^3 \in \mathbb{C}^{1\times 200}$. It can be seen that the size of data to be archived reduces from 30×200 to 3×200.

Because optimum phase series of kth sequence is relative to their individual reference [Fig. 2(a)], phase connection in step 13 in Table II is used to combine all optimal phase series to a common reference. According to Ansari *et al.* [1], the correction to the same reference can be achieved by performing refined EMI on the combined data set $[\hat{z}_{\text{cal},i}^1; \hat{z}_{\text{cal},i}^2; \hat{z}_{\text{cal},i}^3]$ and $[\hat{z}_{\text{cal},i+1}^1; \hat{z}_{\text{cal},i+1}^2; \hat{z}_{\text{cal},i+1}^3]$ [Fig. 2(b)]. Then, corrected phase series relative to a unique reference are obtained by $\hat{\theta}^i := \hat{\theta}^i + \hat{\theta}_{\text{cal}}^i$, where $\hat{\theta}_{\text{cal}}^i$ is a calibration phase constant corresponding to the ith phase series. The procedure of phase connection can be regarded as a subspace clustering by the block diagonal representation [31], whose goal is to group the compressed signals from linear subspaces into a unique subspace.

In the multiple data compression techniques [32], [33], it is assumed that the compressed data are drawn from the multiple low rank subspaces. Therefore, the dimension number of these subspaces is the unique parameter that needs to be determined. In the sequential estimator [1], this parameter is predetermined to 1 as it represents the most coherent scatter mechanism. The isolated scatter mechanism of mini-stacks is represented by linear projections of different orthogonal basis. These orthogonal bases ensure that subspaces of which

TABLE III

PSEUDOCODE OF SEQUENTIAL ESTIMATOR FOR ESD ESTIMATION WHEN NEW IMAGES ARE ADDED

When new images are added

Fill $s-(m-s\times K)$ images to the $K+1^{th}$ mini-stack. Remaining images are separated into k_1 isolated mini-stacks if necessary.

for $k = K+1$ *to* $K+k_1+1$

Input data: $\hat{\mathbf{g}}_i^k = [\tilde{g}_i^1; \ldots; \tilde{g}_i^{k-1}; \mathbf{g}_i^k]$, $\hat{\mathbf{g}}_{i+1}^k = [\tilde{g}_{i+1}^1; \ldots; \tilde{g}_{i+1}^{k-1}; \mathbf{g}_{i+1}^k]$

TABLE II *6* to*10*

end

Phase Connection: $[\hat{\theta}_{cal,i}^1, \hat{\theta}_{cal,i}^2, \ldots, \hat{\theta}_{cal,i}^{K+k_1+1}]$, $[\hat{\theta}_{cal,i+1}^1, \hat{\theta}_{cal,i+1}^2, \ldots, \hat{\theta}_{cal,i+1}^{K+k_1+1}]$

Connected Phase Series: $[\hat{\theta}_i^1, \hat{\theta}_i^2, \ldots, \hat{\theta}_i^{K+k_1+1}]$, $[\hat{\theta}_{i+1}^1, \hat{\theta}_{i+1}^2, \ldots, \hat{\theta}_{i+1}^{K+k_1+1}]$

ESD Phase Series: Differencing Connected Phase Series by Eq.(1)

Mis-registration Estimation Eq. (2) $\rightarrow$ $[\widehat{\Delta x}^1, \widehat{\Delta x}^2, \ldots, \widehat{\Delta x}^{K+k_1+1}]$

principle axes are generated individually can be grouped [34]. The procedure discovering various clusters of data among the different subspaces is named the subspace clustering. During phase connection, the prior cluster number is set to 1, which represents the phase series in different subspaces are grouped into the unique time domain. Accordingly, this problem is equivalent to the standard PCA as the number of clusters is reduced to 1 [35]. Because the recently developed maximum-likelihood estimator EMI [21] is superior to PCA [29], sequential estimator with EMI can guarantee a better quality of phase time series. After the phase calibration in Fig. 2(c), the ESD phase and misregistration values can be calculated by (1) and (2), respectively.

It is worth noting that the remaining mini-stack with size $m - K \times s$ is not compressed. When new acquisitions are included, $s - (m - K \times s)$ images are supplemented to the $K+1$th mini-stack. Then, the estimation procedure is the same as Table II. Table III presents pseudocode for new SLC series processing.

As illustrated in Section II-B, if the stack size is larger than the number of homogeneous pixels, then the determinant of the sample coherence matrix will be 0. This means coherence matrix is not full rank matrix and its inverse $\hat{\Gamma}^{-1}$ will amplify coherence error, leading to the misestimation of phase series. Sequential estimator and double sample in the overlap region can help to mitigate this issue by decreasing stack size and simultaneously increasing the number of homogeneous pixels. This becomes a main advantage of our method and better than regularizing the coherence matrix by heuristically adding a small quantity to its diagonal.

D. Efficient ESD for Azimuth Co-Registration

Fig. 3 summarizes the flowchart of efficient ESD (EESD) by combining refined EMI and sequential estimator. After geometrical co-registration with external DEM, the overlap regions are stacked for homogeneous pixel selection using FaSHPS algorithm, followed by phase optimization in Table II burst by burst. Then, the ESD phase is estimated by differentiating optimum phase using (1), followed by a search through the solution space using (2). When new acquisitions are added, the phase connection is carried out on the new images and the archived data without the necessity

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

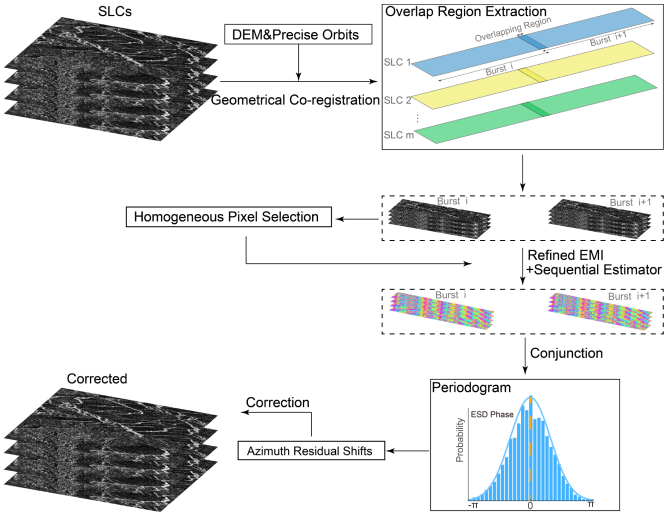


Fig. 3. Workflow of EESD.

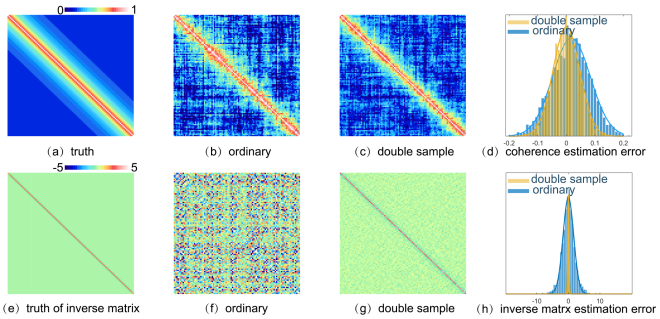


Fig. 4. Simulated coherence and its inverse matrix (homogeneous pixel number is set to 50). (a) True coherence. (b) Coherence estimated by ordinary sample. (c) Coherence estimated by double sample in the overlap region. (d) Histograms of the estimate error for (b) and (c). (e)–(g) Inverse matrix of (a)–(c). (h) Histograms of the difference between estimation in (f) [(g)] and the truth in (e).

for reprocessing the entire stack (Table III). This step is less time-consuming than previous time-series co-registration approaches that use all data set for the least-squares adjustment.

III. Performance Assessment: Synthetic Data

In this section, we carry out comprehensive comparisons of proposed method against the state-of-the-art techniques, including NESD, MST-ESD, joint estimator, and EESD in Table I.

Similar to the previous test in [1] and [21], we employ an exponential decorrelation model to simulate the representative low coherence scenarios with the constant time of 27 days, a short-term coherence of 0.6 and a long-term coherence of 0.1. We set the revisit cycle of $m = 100$ images to six days and the homogeneous pixel number is $n = 50$.

A. Comparisons of Phase Estimators

Fig. 4 presents a realization of sample coherence matrix and its inverse. In contrast to Fig. 4(b), one can see the bias mitigation in Fig. 4(c). The histogram in Fig. 4(d) clearly demonstrates that the dispersion is smaller for coherence

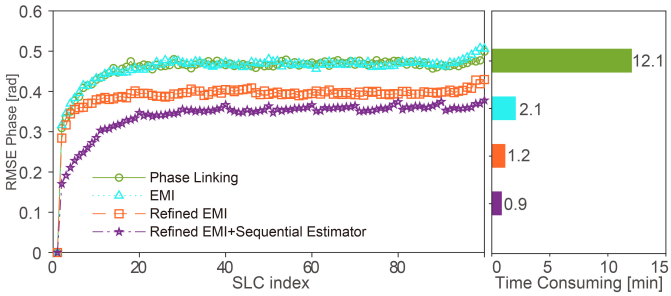


Fig. 5. RMSEs of PL, EMI, refined EMI, and refined EMI + sequential estimator using 4000 times Monte Carlo simulation.

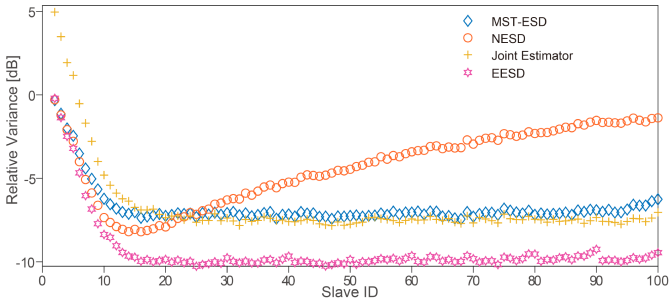


Fig. 6. Relative variances of MST-ESD, joint estimator, NESD, and EESD with respect to the single-master estimator. The first image is chosen as the master reference.

estimated using double sample. Quantitatively, the variance and sample mean reduce from 0.0049 to 0.0004 and 0.0427 to 0.0186, respectively, before and after using double sample, showing improved precision and accuracy. Moreover, it is clear that the coherence error affects the inverse in Fig. 4(h). Since the coherence matrix in Fig. 4(b) is not positive and definite when $n < m$, the coherence error is amplified in Fig. 4(f) compared with Fig. 4(g).

We further use different estimation in Fig. 4 to optimize the phase, as shown in Fig. 5. As expected, refined EMI using double sample can obtain better accuracy than conventional EMI, whereas integrating refined EMI and sequential estimator further reduce the root-mean-square error (RMSE). We can attribute the increase of accuracy to the sufficiency of sample coherence matrix close to truth. From the computational viewpoint, PL is a computer-intensive technique and the burden extremely relies on the iterative number and initial value. For EMI, the regularization is needed to correct the negative definiteness of coherence matrix. The run time is reduced in refined EMI. The sequential estimator further reduces the costs by dividing the whole stack into the smaller blocks. This case demonstrates the effectiveness of our method for phase optimization.

B. Co-Registration Performance

Fig. 6 presents the accuracy of azimuth shifts for different ESD estimators. The relative variances with respect to the conventional single-master ESD estimator are transformed to dB. The smaller value indicates higher accuracy. It is clear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

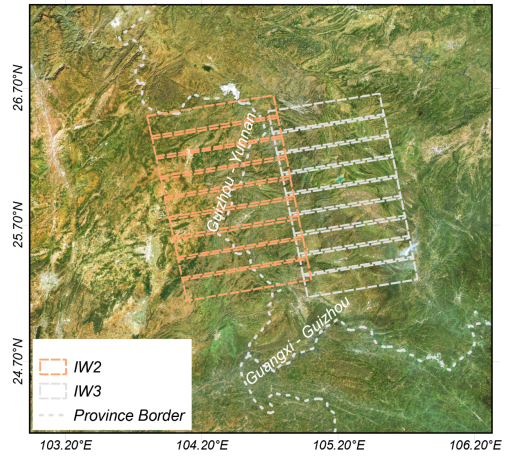


Fig. 7. Footprint of Sentinel-1 TOPS data (track 164).

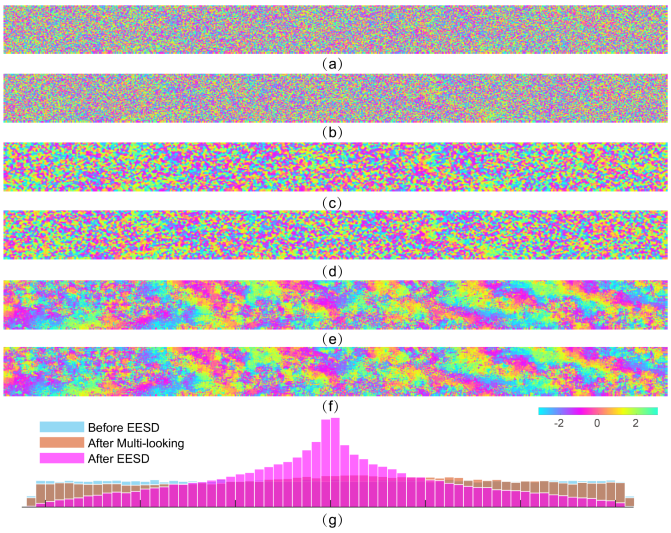


Fig. 8. Zoomed-in view of burst overlap interferograms and corresponding histograms of the ESD phase (the sixth overlap region of IW2 in 20170904 and 20190602). (a) and (b) Upper and lower original interferograms in the overlap region, respectively. (c) and (d) Interferograms after multilooking with 9 × 9 boxcar window. (e) and (f) Interferograms after phase optimization. The blue bar, gray bar, and purple bar in (g) represent the ESD phase calculated by differentiating (a) and (b), (c) and (d), and (e) and (f), respectively.

that EESD outperforms other methods when using the best phase observations in each burst interferogram.

IV. Performance Assessment: Real Data

A total of 54 ascending Sentinel-1 TOPS mode images acquired from track 164 between September 2017 and June 2019 are used in this study. The study area is located at the north edge of the Yunan–Kweichow plateau, China, with a total area of approximately 250 km × 160 km. Heavily influenced by humid subtropical climate, the rainfall volumes in the study area range from 600 to 2000 mm/a. The most regions in this mountainous zone with elevations between 3000 and 5000 m are covered by heavy vegetation in Fig. 7. Such characteristics are likely to cause the fast decorrelation and challenge the capability of the azimuth co-registration.

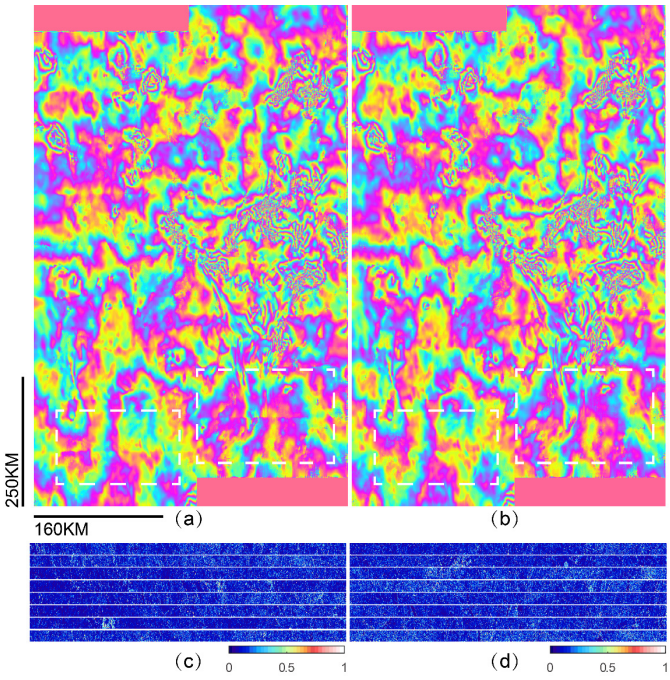


Fig. 9. Interferograms (20180806–20180830) with and without applying EESD. (a) Without EESD. (b) With EESD. Interferograms are multilooked by 20 × 5 looks (azimuth × range) and filtered by Goldstein approach via 128 × 128 boxcar kernel. (c) and (d) Coherence maps over the overlap regions in IW2 and IW3, respectively.

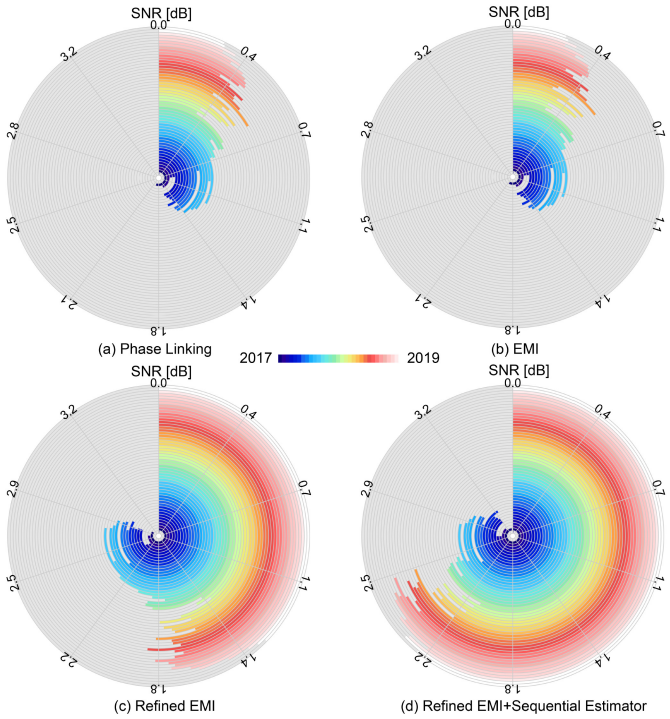


Fig. 10. Radial bar chart of SNR for four phase estimators. (a) PL. (b) EMI. (c) Refined EMI. (d) Refined EMI + sequential estimator.

A. Visual Inspection

Fig. 8 presents a comparison of the two burst interferograms in the overlap region before and after phase optimization. The results estimated from 9 × 9 boxcar window are also shown. The fringe patterns that cannot be observed in Fig. 8(a)–(d)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

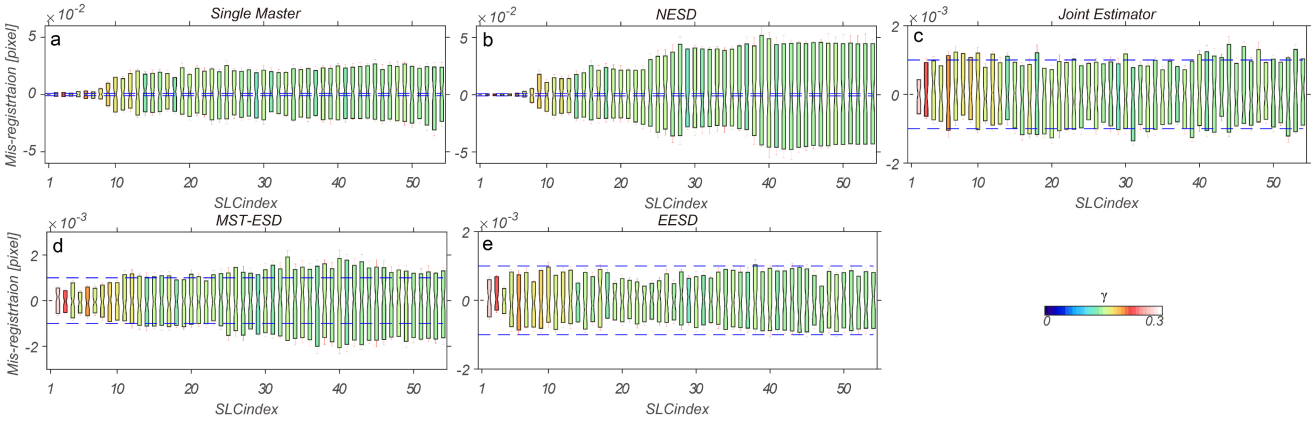


Fig. 11. Boxplot of the residual azimuth shift time series. (a) Single Master. (b) NESD. (c) Joint Estimator. (d) MST-ESD. (e) EESD. The blue lines denote the required −0.001 and 0.001 mis-registration pixel. The color indicates the average coherence between each slave image and common master image in the overlap region.

can be clearly identified in Fig. 8(e) and (f). This leads to a significant peak in the ESD phase histogram in Fig. 8(g).

After estimating all azimuth shifts for two subswaths, Fig. 9 shows the co-registered interferograms before and after EESD. We can see that the phase inconsistence which arises at the bottom of the image in Fig. 9(a) has been eliminated in Fig. 9(b). Multilooking and Goldstein filtering [36] has been performed on this low coherence scenario for better presentation. Coherence maps of all overlap regions for this pair are displayed in Fig. 9(c) and (d), respectively.

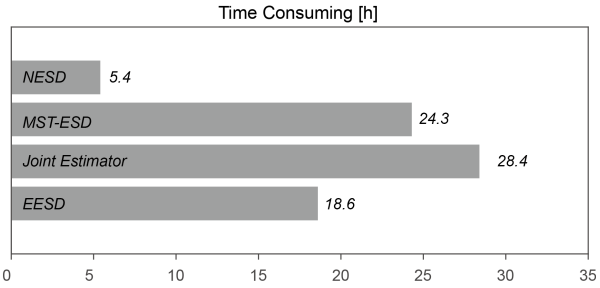


Fig. 12. Run time of different ESD estimators.

B. Quantitative Assessment

We first evaluate the accuracy of four phase estimators using the following SNR:

$$K = 10\log_{10}\frac{\sigma^2_{\phi_{\mathrm{ESD}}}}{\sigma^2_{\hat{\phi}_{\mathrm{ESD}}}} \tag{7}$$

where $\sigma^2_{\phi_{\mathrm{ESD}}}$ and $\sigma^2_{\hat{\phi}_{\mathrm{ESD}}}$, respectively, represent the sample variance of each ESD phase before and after phase optimization, and larger value of K denotes the better performance. The results in Fig. 10 show that refined EMI apparently improves the quality of ESD phase for the time series, while adding the sequential estimator further enhances SNR. This result agrees well with the simulation in Fig. 5 and confirms the effectiveness of the proposed method.

We further evaluate the co-registration accuracy by counting residual azimuth shifts for $m-1$ slave images (the first is set to be master image), which are calculated by means of the resampled time-series images in the previous co-registration. If the co-registration accuracy is high enough, the residual azimuth shifts in the next iteration should be close to 0 for all acquisitions. We repeated the co-registration 50 times under each method and then plotted the residual distribution in Fig. 11. For such low coherence scenario, the conventional multilooking causes artifacts over strong noise areas and results in larger residual shifts for the single master co-registration and NESD. The largest residual shift on NESD demonstrates that the least-squares adjustment may increase co-registration error due to error propagation over very low coherence scenes. On the contrary, Fig. 11(b)–(d) demonstrates the importance of phase quality on the TOPS co-registration. Several co-registration errors larger than 0.001 pixels in MST-ESD and the joint estimator are corrected in Fig. 11(d), showing the robustness of EESD in such case.

C. Computational Efficiency

The run time shown in Fig. 12 is counted on a desktop computer equipped with an Intel Core I7 processor (3.2 GHz) and 16-GB RAM. The implementation is in MATLAB 2018b software. NESD takes about 5.4 h for a 13 115 × 48 518 × 54 (azimuth × range × acquisitions) data volume, while the time increases to 24.3 h for MST-ESD due to an additional burden on coherence estimation and the least-squares adjustment. Note that joint estimator is the most time-consuming method which takes about 28.4 h. The iterative convergency process is responsible for increased burden. Our EESD method takes a total of 18.6 h. Compared with the joint estimator that employs the PL technique, EMI and single-master configuration help save much time. Therefore, we can conclude that EESD is an efficient estimator that can balance the TOPS co-registration accuracy and computational efficiency.

V. Conclusion

In this article, we investigated the sequential TOPS co-registration for time-series Sentinel-1 data over low coherence scenarios by the EESD approach. The incentive behind

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

this method is to improve the quality of the ESD phase. Instead of using the PL technique, we employed EMI to increase the computational efficiency. Taking double sample in the overlap region into account, we further improved the estimate accuracy of covariance matrix and, therefore, the performance of EMI. On the other hand, by invoking data compression techniques, the new acquisitions can be processed without the necessity for reprocessing the entire stack. The effectiveness of presented method against the state-of-the-art estimators has been validated with the synthetic and real data. The comprehensive assessments confirm the outstanding performance of EESD in both estimation and computational efficiencies. Therefore, we can conclude that EESD can be expected to be useful for big SAR data analysis over low coherence scenes.

ACKNOWLEDGMENT

The authors would like to thank four anonymous reviewers for their helpful comments. The Sentinel-1 data were provided by ESA/Copernicus. The authors would also like to thank Dr. Homa Ansari from the German Aerospace Center (DLR), Oberpfaffenhofen, Germany, and Dr. Chang-Jun Zhao from the Chinese Academy of Sciences (RADI), Beijing, China, for the fruitful discussions. The source code of coherence estimation and FaSHPS algorithm used in this article is available from http://mijiang.org.cn/index.php/software/.

REFERENCES

[1] H. Ansari, F. De Zan, and R. Bamler, "Sequential estimator: Toward efficient InSAR time series analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5637–5652, Oct. 2017.

[2] R. Torres *et al.*, "GMES Sentinel-1 mission," *Remote Sens. Environ.*, vol. 120, pp. 9–24, May 2012.

[3] F. De Zan and A. Monti Guarnieri, "TOPSAR: Terrain observation by progressive scans," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 9, pp. 2352–2360, Sep. 2006.

[4] A. Bertran Ortiz and H. Zebker, "ScanSAR-to-stripmap mode interferometry processing using ENVISAT/ASAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3468–3480, Nov. 2007.

[5] N. Yague-Martinez *et al.*, "Interferometric processing of Sentinel-1 TOPS data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2220–2234, Apr. 2016.

[6] E. Sansosti, P. Berardino, M. Manunta, F. Serafino, and G. Fornaro, "Geometrical SAR image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2861–2870, Oct. 2006.

[7] D. O. Nitti, R. F. Hanssen, A. Refice, F. Bovenga, and R. Nutricato, "Impact of DEM-assisted coregistration on high-resolution SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 1127–1143, Mar. 2011.

[8] P. Prats-Iraola *et al.*, "Role of the orbital tube in interferometric spaceborne SAR missions," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1486–1490, Jul. 2015.

[9] H. Fattahi, P. Agram, and M. Simons, "A network-based enhanced spectral diversity approach for TOPS time-series analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 777–786, Feb. 2017.

[10] R. Scheiber, M. Jager, P. Prats-Iraola, F. De Zan, and D. Geudtner, "Speckle tracking and interferometric processing of TerraSAR-X TOPS data for mapping nonstationary scenarios," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 4, pp. 1709–1720, Apr. 2015.

[11] B. Xu *et al.*, "Continent-wide 2-D co-seismic deformation of the 2015 mw 8.3 Illapel, Chile earthquake derived from Sentinel-1A data: Correction of azimuth co-registration error," *Remote Sens.*, vol. 8, no. 5, p. 376, May 2016.

[12] F. De Zan, P. Prats-Iraola, R. Scheiber, and A. Rucci, "Interferometry with TOPS: Coregistration and azimuth shifts," in *Proc. EUSAR 10th Eur. Conf. Synth. Aperture Radar*, Jun. 2014, pp. 1–4.

[13] P. Prats-Iraola, R. Scheiber, L. Marotti, S. Wollstadt, and A. Reigber, "TOPS interferometry with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 8, pp. 3179–3188, Aug. 2012.

[14] R. Scheiber and A. Moreira, "Coregistration of interferometric SAR images using spectral diversity," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 5, pp. 2179–2191, Sep. 2000.

[15] R. Bamler and M. Eineder, "Accuracy of differential shift estimation by correlation and split-bandwidth interferometry for wideband and Delta-k SAR systems," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 151–155, Apr. 2005.

[16] N. B. D. Bechor and H. A. Zebker, "Measuring two-dimensional movements using a single InSAR pair," *Geophys. Res. Lett.*, vol. 33, no. 16, 2006, Art. no. L16311, doi: 10.1029/2006GL026883.

[17] R. L. Graham and P. Hell, "On the history of the minimum spanning tree problem," *IEEE Ann. Hist. Comput.*, vol. 7, no. 1, pp. 43–57, Jan. 1985.

[18] N. Yague-Martinez, F. De Zan, and P. Prats-Iraola, "Coregistration of interferometric stacks of Sentinel-1 TOPS data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1002–1006, Jul. 2017.

[19] A. M. Guarnieri and S. Tebaldini, "On the exploitation of target statistics for SAR interferometry applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3436–3443, Nov. 2008.

[20] F. De Zan, M. Zonno, and P. Lopez-Dekker, "Phase inconsistencies and multiple scattering in SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6608–6616, Dec. 2015.

[21] H. Ansari, F. De Zan, and R. Bamler, "Efficient phase estimation for interferogram stacks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4109–4125, Jul. 2018.

[22] Z.-F. Ma, M. Jiang, Y. Zhao, R. Malhotra, and B. Yong, "Minimum spanning tree co-registration approach for time-series Sentinel-1 TOPS data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3004–3013, Aug. 2019.

[23] M. Jiang, X. Ding, R. F. Hanssen, R. Malhotra, and L. Chang, "Fast statistically homogeneous pixel selection for covariance matrix estimation for multitemporal InSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1213–1224, Mar. 2015.

[24] M. Jiang, X. Ding, and Z. Li, "Hybrid approach for unbiased coherence estimation for multitemporal InSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2459–2473, May 2014.

[25] M. Jiang and A. M. Guarnieri, "Distributed scatterer interferometry with the refinement of spatiotemporal coherence," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3977–3987, Jun. 2020.

[26] M. Jiang, X. Ding, Z. Li, X. Tian, C. Wang, and W. Zhu, "InSAR coherence estimation for small data sets and its impact on temporal decorrelation extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6584–6596, Oct. 2014.

[27] C.-A. Deledalle, L. Denis, F. Tupin, A. Reigber, and M. Jager, "NL-SAR: A unified nonlocal framework for resolution-preserving (Pol)(In)SAR denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2021–2038, Apr. 2015.

[28] M. E. Wall, A. Rechtsteiner, and L. M. Rocha, "Singular value decomposition and principal component analysis," in *A Practical Approach to Microarray Data Analysis*, D. P. Berrar, W. Dubitzky, M. Granzow, Eds. Boston, MA, USA: Springer, 2003, pp. 91–109.

[29] G. Fornaro, S. Verde, D. Reale, and A. Pauciullo, "CAESAR: An approach based on covariance matrix decomposition to improve multibaseline–multitemporal interferometric SAR processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2050–2065, Apr. 2015.

[30] X. X. Zhu and R. Bamler, "Let's do the time warp: Multicomponent nonlinear motion estimation in differential SAR tomography," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 735–739, Jul. 2011.

[31] C. Lu, J. Feng, Z. Lin, T. Mei, and S. Yan, "Subspace clustering by block diagonal representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 487–501, Feb. 2019.

[32] J. E. Fowler, "Compressive-projection principal component analysis," *IEEE Trans. Image Process.*, vol. 18, no. 10, pp. 2230–2242, Oct. 2009.

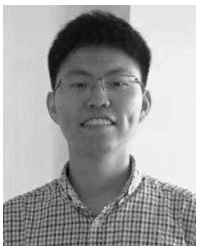
[33] A. K. Jain, "Image data compression: A review," *Proc. IEEE*, vol. 69, no. 3, pp. 349–389, Mar. 1981.

[34] E. Elhamifar and R. Vidal, "Sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 2790–2797.

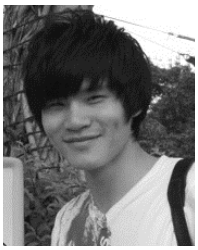
[35] R. Vidal, "Subspace clustering," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 52–68, Mar. 2011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.

[36] M. Jiang, X. Ding, X. Tian, R. Malhotra, and W. Kong, "A hybrid method for optimization of the adaptive goldstein filter," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 29–43, Dec. 2014.



Zhang-Feng Ma (Student Member, IEEE) was born in Nantong, China, in 1995. He received the B.S. degree in geomatics engineering from the Nanjing University of Technology, Nanjing, China, in 2017, and the M.Sc. degree in geomatics engineering from Hohai University, Nanjing, in 2019, where he is pursuing the Ph.D. degree. His research interests include interferometric synthetic aperture radar and its application for monitoring and modeling crustal deformation.



Mi Jiang (Senior Member, IEEE) was born in Nanjing, China, in 1982. He received the B.S. degree in computational mathematics from the Nanjing University of Technology, Nanjing, in 2005, the M.Sc. degree in radar remote sensing from Central South University, Changsha, China, in 2009, and the Ph.D. degree from The Hong Kong Polytechnic University, Hong Kong, in 2014.

He is a Professor with the School of Earth Sciences and Engineering, Hohai University, Nanjing. His research interests include the field of statistical inference with an emphasis on interferometric synthetic aperture radar processing.



Teng Huang was born in 1957. He received the B.S. degree in water conservancy engineering and the M.Sc. degree in geomatics engineering from the East China Institute of Water Resources (now Hohai University), Nanjing, China, in 1980 and 1995, respectively.

He is engaged in the teaching and scientific research in surveying and mapping for 30 years. He has authored more than 70 academic articles and 3 books, and supervised more than 70 master and doctoral students. His research interests include precision engineering survey and large engineering safety monitoring theory and its application research.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:43:33 UTC from IEEE Xplore. Restrictions apply.